

INDEPENDENT PEER-REVIEW PACKET

EC-018

**Full-Dimensional Interval-Certified Degree-Four
Route-B Box**

Cold Review Packet 13 of 20

Dylan M. Roy Mathematical Research Program

Prepared for independent validation - 27 July 2026

This packet is self-contained for mathematical review. Drive access is supplemental, not required to understand the claim and proof.

READ FIRST

You are being asked to review one narrow terminal object, not the wider q0/P0.2/Theorem B research program. The declared terminal label is an internal research-state classification. Your task is to decide whether the exact statement and its proof/evidence are correct within the stated hypotheses and exclusions.

1. Pass A - Independent reconstruction. Read the exact statement, notation, and hypotheses. Attempt your own derivation or falsification before reading the embedded historical review record.
2. Pass B - Source audit. Compare your reconstruction with the complete controlling source text embedded in Part II. Check every dependency, sign, factor, quantifier, scope boundary, and computational claim.
3. Pass C - Hostile review. Run the object-specific focal checks and invent at least one additional mutation or counterexample attempt.
4. Issue one fail-closed verdict: APPROVE; APPROVE WITH CLARIFICATIONS; AMEND REQUIRED; REJECT; or CANNOT VERIFY.

Do not transfer approval to excluded downstream claims. Do not count access to this packet, the author, or another reviewer as independent mathematical evidence.

Part I - Exact Review Object

Packet	13 of 20
Object ID	EC-018
Object class	Executable interval certificate
Exact statement	At $r=1/20$, every point in P01-ROUTE-B-D4-BOX-001 satisfies the repaired degree-four capture sufficient conditions.
Internal terminal label	CLOSED - EXACT FULL-DIMENSIONAL INTERVAL-CERTIFIED DEGREE-FOUR ROUTE-B BOX AT $r=1/20$
Terminal date	2026-07-22
Recommended expertise	Validated numerics / interval arithmetic / dynamical systems

1. Significance

This object was separated from larger theorem claims so that its correctness can be assessed without accepting unrelated Gaussian-field, persistence, adjacency, or asymptotic assertions. A positive review supports only the exact statement above and the dependency uses explicitly listed below.

2. Evidence class and prior review

Evidence	Byte-exact source capsule, full nine-dimensional interval certificate, all margins positive, and five falsification controls.
Prior review record	Independent clean-environment execution reproduced the source hashes, verdict, minimum margin, and controls.
Binding exclusions	No Gaussian/Palm mass, exact-field transfer, r -uniformity, P0.1/P0.2, or theorem conclusion.

3. Dependency map and use restrictions

- Inputs: only definitions, hypotheses, and subordinate identities explicitly reproduced in the controlling source text.
- Output: only the exact statement listed in the review-object table.
- Permitted downstream use: as a cited lemma/interface under identical object definitions and hypotheses.
- Forbidden transfer: no approval of a parent theorem, numerical constant, external release, or canonical promotion unless separately reviewed.

- Object-specific exclusions: No Gaussian/Palm mass, exact-field transfer, r-uniformity, P0.1/P0.2, or theorem conclusion.

4. Object-specific hostile checks

- Decode and hash the embedded source before execution.
- Rerun in a clean arbitrary-precision environment and reproduce all margins.
- Confirm each negative control fails the intended gate rather than a different accidental gate.
- Attempt at least one additional mutation that changes an orientation, sign, normalization, law, endpoint, or quantifier while leaving the text superficially similar.

5. Proof-obligation ledger

Obligation	Reviewer test	Initial state
O1 Object identity	Definitions, ordering, normalization, and law match the statement.	OPEN FOR REVIEW
O2 Hypotheses	Every hypothesis used is stated and no stronger premise is hidden.	OPEN FOR REVIEW
O3 Derivation	Each algebraic, analytic, topological, or probabilistic step is valid.	OPEN FOR REVIEW
O4 Quantifiers	Uniformity, existence, endpoint, and asymptotic quantifiers are correct.	OPEN FOR REVIEW
O5 Computation	Any executable or interval result is reproducible and non-vacuous.	OPEN FOR REVIEW
O6 Scope firewall	The conclusion is not silently enlarged beyond the exact object.	OPEN FOR REVIEW
O7 Contradictions	No known correction, counterexample, or retained caveat defeats the claim.	OPEN FOR REVIEW

Part II - Complete Controlling Source Text

The following is a paragraph-order-preserving text reproduction of the controlling Drive artifact. Typography is normalized for this packet; mathematical content is not intentionally rewritten. Use the manifest URL for native-file or byte-level checks.

GP-DATA-106-v1.0 — BYTE-EXACT SOURCE CAPSULE

Full-Dimensional Route-B Degree-Four Box

Artifact ID: GP-DATA-106-v1.0

Class: DATA — gzip/base64 source and receipt capsule

Authority: none

Canonical impact: NONE

SOURCE

bytes: 8709

sha256: 2eb38f92c83dee204d248357353d02770b076fd58bc6293e8f72abfb6ddf6c2e

gzip bytes: 2989

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END_RECEIPT_GZIP_BASE64

Status: SOURCE CAPSULE PUBLISHED — FRESH DRIVE RECONSTRUCTION REQUIRED BEFORE USE

Part III - Terminal-State Reconciliation

1. Closure decision

Exact internal disposition	CLOSED - EXACT FULL-DIMENSIONAL INTERVAL-CERTIFIED DEGREE-FOUR ROUTE-B BOX AT $r=1/20$
Closure date	2026-07-22
Evidence basis	Byte-exact source capsule, full nine-dimensional interval certificate, all margins positive, and five falsification controls.
Independent review already recorded	Independent clean-environment execution reproduced the source hashes, verdict, minimum margin, and controls.

The prior closure record is evidence about the research program's internal state, not a substitute for your review. A new material objection should be reported even if the internal label is CLOSED.

2. Correction and reopening rule

Reopen or amend this object if you identify a counterexample, an object-definition mismatch, a sign/factor/quantifier defect, an unreproducible decisive computation, a hidden premise, or a scope claim not supported by the proof. Corrections must be additive and must preserve the historical source and review record.

3. Limits

- No Gaussian/Palm mass, exact-field transfer, r -uniformity, P0.1/P0.2, or theorem conclusion.
- A correct lemma may still be unusable downstream if another dependency uses a different field, law, coordinate convention, branch mark, normalization, or parameter domain.
- A prior independent review does not reduce your duty to reconstruct the argument; duplicate lineages must not be counted as additional independence.

Part IV - Reproduction Instructions

- Record your name, institution or affiliation, expertise, date, and any prior exposure to this research program.
- Reconstruct the exact statement from Part I without relying on the historical verdict.
- Audit Part II line by line. For computational objects, obtain the raw carrier from the manifest, verify hashes first, inspect source safety, then execute in a clean environment.
- Complete the obligation ledger and hostile checks. Preserve exact counterexamples, code, logs, and environment details.
- Return the verdict form below plus a finding list with severity: critical, material, clarification, editorial, or no issue.

Part V - Reviewer Verdict Form

Reviewer	_____
Affiliation / expertise	_____
Prior exposure	None / limited / substantial - explain: _____
Independent reconstruction	Complete / partial / not possible
Verdict	APPROVE / APPROVE WITH CLARIFICATIONS / AMEND REQUIRED / REJECT / CANNOT VERIFY
Exact scope accepted	_____
Critical or material findings	_____ _____

Reproduction artifacts

Code/log/hash/location: _____

Signature and date

Required narrative: summarize your independent derivation; list every disagreement; state whether each issue affects the theorem statement, only the proof, only reviewability/reproducibility, or only presentation. State explicitly which excluded claims you did not review.

Part VI - Manifest and Provenance

Controlling source: <https://docs.google.com/document/d/1sAbtS40ioX91gxiYRpbTZVg2hmqDqQu-z5HklYMibe4/edit>

Closure / terminal record: https://docs.google.com/document/d/1-WDQtU4LpHneNztZss6ctg_IKhsyGnXdYC4Y7wM6Fc/edit

Local source snapshot used to construct this packet: GP-DATA-106-v1.0 — Byte-Exact Capsule for the Full-Dimensional Route-B Degree-Four Box

Local source snapshot SHA-256 (DOCX export, transport checksum only):
59ca40be7301607a5c5b30cf93c6cb94674e9772a74b45c0a7c4364cc0faef5d

Authority note: the transport checksum above identifies the locally downloaded Office export used for packet construction. It is not substituted for any marker-delimited frozen-body hash declared inside the source artifact.

Supplemental Drive access may be granted for raw capsules, revision history, comments, exact native exports, and dependency artifacts. The packet is designed so that those materials strengthen reproducibility rather than supply missing mathematical context.